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The functional biology of human milk oligosaccharides

Lars Bode *

Department of Pediatrics, University of California, San Diego, 9500 Gilman Dr., 0715, La Jolla, CA 92093, USA

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ABSTRACT

Human milk oligosaccharides (HMOs) are a group of complex sugars that are highly abundant in human milk, but currently not present in infant formula. More than a hundred different HMOs have been identified so far. The amount and composition of HMOs are highly variable between women, and each structurally defined HMO might have a distinct functionality. HMOs are not digested by the infant and serve as metabolic substrates for select microbes, contributing to shape the infant gut microbiome. HMOs act as soluble decoy receptors that block the attachment of viral, bacterial or protozoan parasite pathogens to epithelial cell surface sugars, which may help prevent infectious diseases in the gut and also the respiratory and urinary tracts. HMOs are also antimicrobials that act as bacteriostatic or bacteriocidal agents. In addition, HMOs alter host epithelial and immune cell responses with potential benefits for the neonate. The article reviews current knowledge as well as future challenges and opportunities related to the functional biology of HMOs.

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1. What are human milk oligosaccharides?

Human milk oligosaccharides (HMOs) are unconjugated complex glycans (sugars and carbohydrates) that are highly abundant in human milk, but not in infant formula [1]. One liter of mature human milk contains 10–15 g HMO, which often exceeds the total amount of protein and is 100- to 1000-fold higher than the concentration of oligosaccharides in bovine milk, which is the basis of most infant formula. Concentrations are even higher in human colostrum.

HMOs consist of five monosaccharide building blocks: galactose (Gal), glucose (Glc), N-acetylglucosamine (GlcNAc), fucose (Fuc) and the sialic acid (Sia) derivative N-acetyl-neuraminic acid. All HMOs carry lactose (Galβ1-4Glc) at the reducing end, which can be elongated in β1-3- or β1-6-linkage by two different disaccharides, either Galβ1-3GlcNAc (type 1 chain) or Galβ1-4GlcNAc (type 2 chain). HMOs with more than 15 disaccharide units have been described, forming complex structural backbones that can be further modified by the addition of Fuc or/and Sia.

1.1. HMO fucosylation is related to Lewis blood group antigens

Fuc can be added to the HMO backbone in α1-2-, α1-3- or α1-4-linkage. HMO fucosylation is highly dependent on the mother's Lewis

* Tel.: +1 858 246 1874.
E-mail address: lbode@ucsd.edu.
URL: http://www.bodelab.com.

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blood group status [2–5]. An enzyme called fucosyltransferase 2 (FUT2) catalyzes the addition of Fuc in α1-2-linkage on Lewis blood group epitopes as well as on HMOs [2]. FUT2 is actively expressed in over 70% of Caucasian women (Secretors). The milk of Secretor women contains high concentrations of α1-2-fucosylated HMOs, *e.g.* 2′-fucosyllactose (2′FL) and lacto-N-fucopentaose 1 (LNFP1). Nonsecretors, however, do not express an active FUT2 and the milk of Nonsecretor women lacks α1-2-fucosylated HMOs like 2′FL or LNFP1.

A separate enzyme called fucosyltransferase 3 (FUT3) catalyzes the addition of Fuc in α1-3/4-linkage (depending on the type of the underlying HMO backbone), and FUT3 can also be inactive in parts of the population (Lewis negative) [3]. The milk of Lewis negative women has markedly reduced concentrations of α1-3/4-fucosylated HMOs.

Depending on the expression of active FUT2 and FUT3 enzymes, women can be separated into four groups: 1. Lewis positive Secretors (FUT2 active, FUT3 active), 2. Lewis negative Secretors (FUT2 active, FUT3 inactive), 3. Lewis positive Nonsecretors (FUT2 inactive, FUT3 active), and 4. Lewis negative Nonsecretors (FUT2 inactive, FUT3 inactive). Accordingly, the oligosaccharide composition in the milk of women from these four groups varies significantly [4,5].

1.2. Variation in HMO sialylation is more subtle than the variation in HMO fucosylation

Sia can be added to the HMO backbone in α2-3- or α2-6-linkage, either to the terminal Gal or to internal GlcNAc. Sia contains a carboxyl-group, which introduces a negative charge to HMOs. Therefore, sialylated (or acidic) HMOs carry one or more negative charges depending on the number of Sia linked to the HMO backbone. Several sialyltransferases are involved in catalyzing the addition of Sia to the HMO backbone. As described above, the fucosyltransferases FUT2 and FUT3 can be inactive and certain fucosylated HMOs can be entirely absent. A complete loss of sialyltransferases and corresponding sialylated HMO has not yet been described. Instead, more variations in HMO sialylation are likely due to subtle interindividual expression variations in sialyltransferases or other enzymes and transporters important in sialylation pathways.

1.3. Genetic and environmental factors that contribute to HMO biosynthesis remain mostly unknown

As outlined for HMO fucosylation, HMO biosynthesis in the human mammary gland is in part genetically determined. Genes other than FUT2 and FUT3 that contribute to chain elongation, branching or sialylation might be differentially expressed in different women and lead to distinct HMO composition profiles. Whether or not other maternal factors like age, diet, general health status, use of medication and drugs, etc. affect HMO biosynthesis remains mostly unknown. Environmental factors that impact HMO composition are currently in the focus of HMO research.

2. HMO metabolism

Once ingested by the breast-fed infant, HMOs resist the low stomach pH as well as degradation by the infant's pancreatic and brush border enzymes [6,7]. Approximately, 1% of the ingested HMOs are absorbed, reach the infant's systemic circulation, and are excreted intact in the infant's urine [8–10]. The majority of HMOs are either metabolized by the infant's gut microbes or excreted intact with the infant's feces [11,12].

Since HMOs are absorbed and appear in the systemic circulation, they likely reach many organs other than the gut, including the liver and the brain, as well as the respiratory and the urinary tract. Thus, the biological functions of HMOs may not be localized to the gut; HMOs might impact the infant on multiple different levels throughout the neonate.

3. HMOs are human milk prebiotics

According to a definition by Roberfroid et al., "a prebiotic is a selectively fermented ingredient that allows specific changes, both in the composition and/or activity in the gastrointestinal microbiota, that confers benefits upon host well-being and health" [13]. HMOs serve as metabolic substrates for specific bacteria like *Bifidobacterium longum* subsp. *infantis*. As a consequence, these bacteria have a growth advantage and thrive. Other bacteria that cannot utilize HMOs have a disadvantage and do not grow as well or not at all. Thus, HMOs are the first prebiotics that humans encounter with their diet, usually from day one of life. A bacteria's ability to utilize HMOs requires an entire set of enzymes, transporters and other molecules. For example, certain bacteria have evolved together with HMOs and express sialidases that cleave Sia and fucosidases that cleave Fuc. Only very few bacteria are capable of degrading the entire set of HMOs [14–16]. Other bacteria may only be able to utilize a limited set of HMOs or specific epitopes on more complex HMOs. For example, bacteria with a certain fucosidase may be able to utilize Fuc, but not Sia. Some bacteria may be able to utilize HMOs only after other bacteria have removed the Fuc or Sia from the backbone, creating a "community feast" where multiple different bacteria may be able to degrade the entire set of HMOs, but only when they act together as a community.

While the sequential degradation of HMOs by different microbes needs to be further elucidated, it is evident that, based on their structural diversity, different HMOs can be metabolized by different bacteria. In other words, not all HMOs lead to the same changes in composition and/or activity in the gastrointestinal microbiota and have the same benefits upon host well-being and health. Prebiotic effects are likely structure-specific, and HMOs are a group of structurally diverse glycans. Since HMO composition varies between women, one can hypothesize that the milk of different women affects the infant gut mircobiome differently, which may relate to short-term infant health outcomes, but also have long-term consequences for health status and disease risk later on in life.

4. HMOs serve as antiadhesives

While the primary focus of HMO research has traditionally been on their prebiotic effects, HMOs are more than just "food for bugs". Many viral, bacterial or protozoan parasite pathogens need to attach to epithelial cell surfaces to proliferate and in some cases invade and cause disease. Often, the initial attachment is to epithelial cell surface sugars (glycans) also known as the glycocalyx. While these glycans are conjugated to proteins or lipids, HMOs resemble some of the glycan structures and serve as soluble decoy receptors that block pathogen binding to epithelial cells. Unbound pathogens can no longer attach to the cell surface and are washed out without causing disease. Norovirus and Rotavirus are examples of viral pathogens that bind to the epithelial glycocalyx; HMOs resemble the glycan binding partners and block viral attachment, providing one explanation for the reduced incidence of these viral infections in breast-fed infants compared to formula-fed infants. *Campylobacter jejuni* [17] and enteropathogenic *E. coli* [18] are examples of bacterial pathogens that follow the same principle and have significant impact on infant health as they are responsible for a majority of bacterial diarrheal episodes.

Our lab has recently shown that HMOs also prevent the attachment of the protozoan parasite *Entamoeba histolytica* [19]. Although uncommon in the US and Europe, *E. histolytica* infects more than 50 million people worldwide and causes the disease amebiasis, leading to more than 100,000 deaths annually [20]. *E. histolytica* expresses a lectin, a glycan-binding protein, which is a major virulence factor involved in *E. histolytica* attachment to intestinal epithelial cells [21,22]. The lectin is also involved in the subsequent killing and phagocytosis of these cells. HMOs block the lectin and prevent attachment, killing and phagocytosis. The effects are structure-specific and require a terminal Gal on

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the HMOs to be effective. Fucosylation of the terminal Gal abolishes the effect, highlighting distinct structure–function relationships of different HMOs.

It is important to emphasize that most of the provided evidence in support of the anti-adhesive effects of HMOs stems from tissue culture attachment assays and animal infection models. Well-designed and fully-powered mother–infant observation studies and, most importantly, intervention studies are required to confirm that an individual HMO or a mixture of different HMOs reduce the incidence of infectious diseases caused by viral, bacterial and protozoan parasite pathogens.

5. HMOs act as antimicrobials

HMOs may protect the neonate from pathogens by acting as prebiotics that provide beneficial bacteria with a growth advantage and by serving as antiadhesives at the interface of microbe–host interactions. In addition, HMOs may have a more direct way of keeping pathogens in check. We have recently shown that *Streptococcus agalacticae* (Group B *Streptococcus*; GBS) is no longer able to proliferate when HMOs are present (Lin et al., manuscript in preparation). GBS is one of the leading neonatal pathogens affecting about 1 in 2000 newborns in the US. An estimated 15–40% of all pregnant women are colonized with GBS in the gastrointestinal or genital tract [23,24]; more than half of them experience miscarriage or stillbirth and 16–53% of colonized mothers may pass the bacteria onto their babies during childbirth [25,26]. Infected infants can develop sepsis, pneumonia, and life-threatening meningitis. GBS colonization in the genital tract also increases the probability of urinary tract infections (UTIs) in pregnant women [25]. We used multidimensional chromatography and identified specific non-sialylated HMOs as having the most pronounced bacteriostatic effect on GBS. We also used a GBS transposon library and identified a GBS mutant that is no longer susceptible to the bacteriostatic effects of HMOs. The mutant lacks a gene that encodes a glycosyltransferase and additional *in vitro* studies suggest that GBS employs the glycosyltransferase to incorporate specific HMOs into their cell membrane, which then stops GBS proliferation similar to some of the commercially available antibiotics. These most recent results are very promising in developing new antibiotics that are based on natural compounds like HMOs and synergize with already available antibiotics that, when used alone, start to lose efficacy due to the development of antibiotic resistance.

HMOs may not only protect from viral, bacterial or protozoan pathogens. Most recent work suggests that HMOs also affect fungal–host interactions [27]. *Candida albicans*, a prevalent fungal colonizer of the neonatal gut [28–30], causes the overwhelming majority of invasive fungal disease in premature infants and is highly associated with life-threatening intestinal disorders like necrotizing enterocolitis and perforation [31,32]. Treatment with HMOs significantly reduced invasion of human premature intestinal epithelial cells (pIECs) by *C. albicans* in a dose dependent manner [27]. The decreased invasive potential of *C. albicans* correlated with a delay in hyphal growth and morphogenesis as well as a reduction in the ability of *C. albicans* to associate with pIECs, processes important for the initial pathogenesis steps of *C. albicans* infections. Again, HMOs appear to directly affect the microbe, here the fungus *C. albicans*, altering hyphal growth and morphogenesis, which then makes it more difficult for the pathogen to attach, invade and cause disease.

6. HMOs alter epithelial and immune cell responses

HMOs may not only impact microbes directly, but also indirectly by altering host cell responses. HMOs have been shown to modulate intestinal epithelial cell apoptosis, proliferation and differentiation [33]. HMOs have also been shown to alter intestinal epithelial cell gene expression leading to changes in the cell surface glycocalyx [34]. Thus, HMOs may not only affect microbe–host attachment by serving as soluble decoy receptors as described above, but also by changing the expression of the glycocalyx receptors by reprogramming the epithelial cell.

We recently investigated whether or not HMOs can serve as anti-adhesives not only for bacteria in the infant gut, but also for bacteria in the urinary tract [35]. After all, HMOs are absorbed and excreted with the urine. Hence, they are present in the urinary tract. To our surprise we found that HMOs indeed reduce invasion of uropathogenic *Escherichia coli* (UPEC), but not by serving as anti-adhesives that reduce UPEC attachment to epithelial cells. Instead, HMOs interact with the epithelial cells and make them more resistant against UPEC attacks. HMOs strongly suppress intracellular signaling of apoptotic pathways that renders the epithelial cell irresponsive when UPEC tries to destroy them. The effects are highly structure-dependent and only sialylated HMOs like 3′-sialyllactose are effective [35].

HMOs not only alter the response of epithelial cells. HMOs also affect immune cells. For example, we have recently shown that specific sialylated HMOs reduce the expression of pro-inflammatory cytokines IL-1β and IL-6 in LPS-stimulated macrophages (Autran et al., manuscript in preparation). This and many other examples from other immune cell types [36,37] suggest that HMOs alter immune responses and contribute to protecting the neonate. Again, HMO composition varies between women, and one can hypothesize that the milk from different women affects the infant's immune system differently, which may contribute to varied risk for the infant to develop allergies, asthma or other disorders related to the immune system.

7. Conclusion

In conclusion, human milk, unlike the milk of most other mammals, contains very high concentrations of a structurally diverse group of more than a hundred different complex sugars called human milk oligosaccharides (HMOs). HMO composition follows a basic blueprint, but each woman produces a distinct profile of different HMOs at different concentrations that can change over the course of lactation. These inter- and intra-individual differences in HMO composition are in part determined by genetics, and associations to environmental and maternal factors remain to be elucidated. HMOs are minimally digested by the infant, but serve as metabolic substrates for specific microbes in the infant gut, shaping the developing gut microbiome as natural prebiotics. HMOs serve as antiadhesives and prevent the attachment of various pathogens to the infant's epithelial surfaces, preventing or reducing infectious diseases in the gut, and potentially also in the respiratory and the urinary tract. HMOs act as antimicrobials and directly inhibit bacterial proliferation with the potential to serve as natural templates in the development of new and desperately needed antibiotics. Moreover, HMOs alter epithelial and immune cell responses with the potential to affect infant's risk to develop allergies, asthma and other disorders.

New technologies for rapid and high-throughput HMO analysis now enable large mother–infant observation studies that investigate associations between HMO composition and various infant health outcomes as well as associations between maternal factors and HMO composition. Advances in HMO synthesis (chemically, enzymatically, biotechnologically alone or in combination) enables mechanistic studies *in vitro* and in suitable animal models and will ultimately allow clinical intervention studies to investigate whether or not HMOs, either alone or in combination, benefit the human neonate.

Conflict of interest statement

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